Programming Fundamentals

LAB - 2

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Roll No. : 46

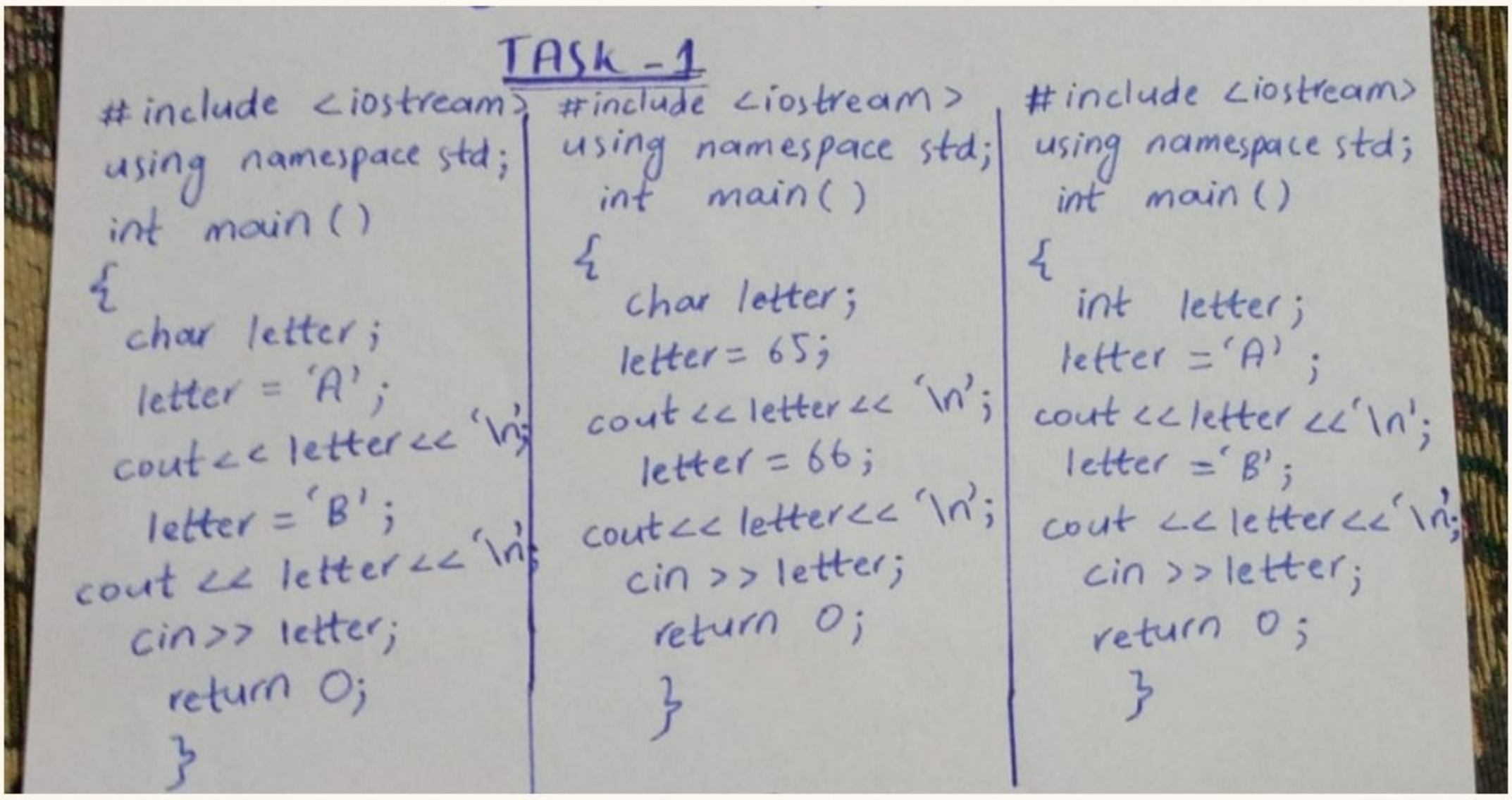
Submitted To : Sir Asad Majeed

Class : CS (C)

Date : 9 oct, 2026.

Lab Task 1

Handwritten code (all three programs)



Typed code, program 1 (char letter, A and B)

```cpp
#include <iostream>
using namespace std;

int main()
{
    char letter;

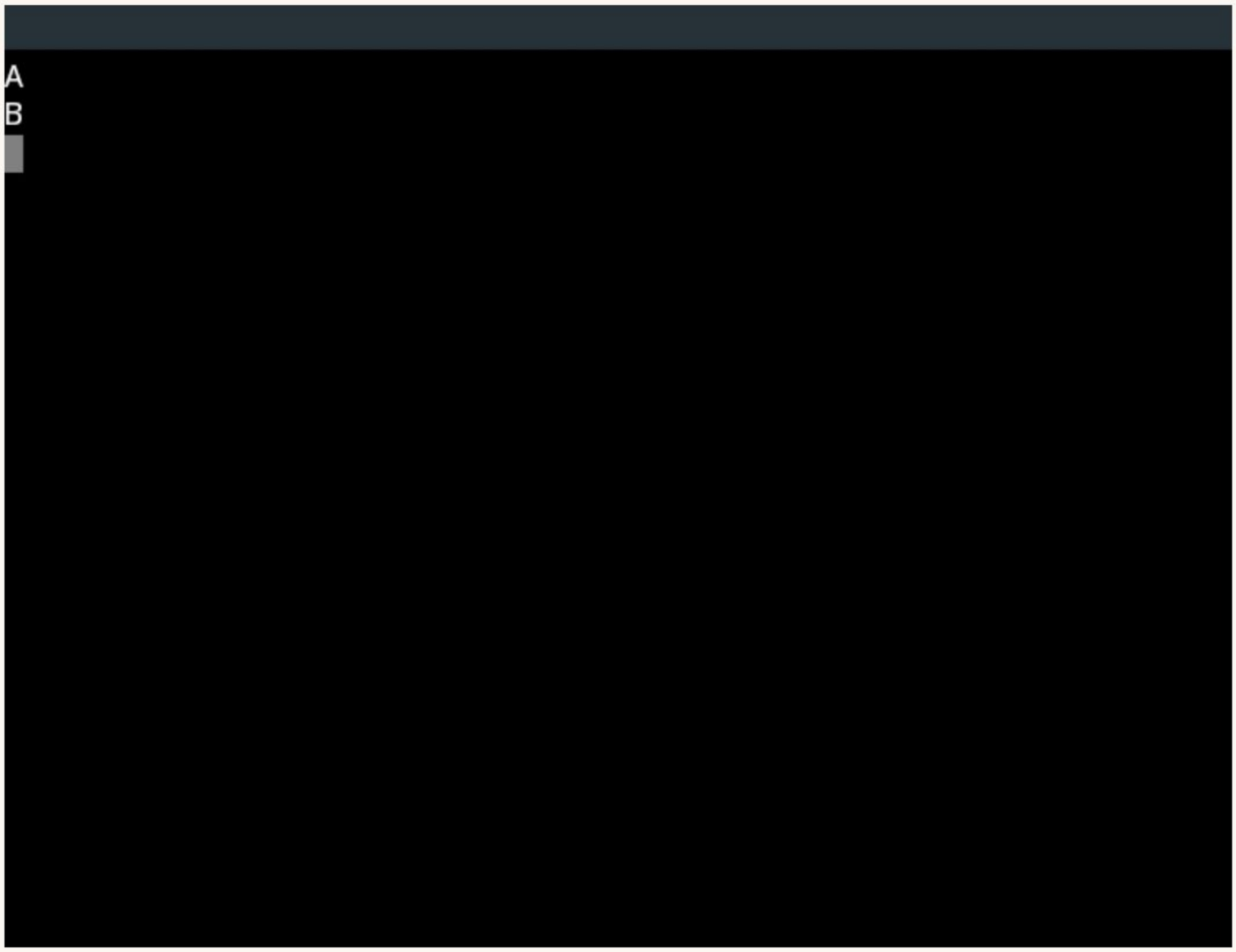
    letter = 'A';
    cout << letter << '\n';

    letter = 'B';
    cout << letter << '\n';

    cin >> letter;

    return 0;
}
```

Output, program 1



Typed code, program 2 (char letter, 65 and 66)

```
#include <iostream>
using namespace std;

int main()
{
    char letter;
    letter = 65;
    cout << letter << '\n';
    letter = 66;
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

Output, program 2

```
A
B
```

Typed code, program 3 (int letter)

```
#include <iostream>
using namespace std;

int main()
{
    int letter;

    letter = 'A';
    cout << letter << '\n';

    letter = 'B';
    cout << letter << '\n';

    cin >> letter;

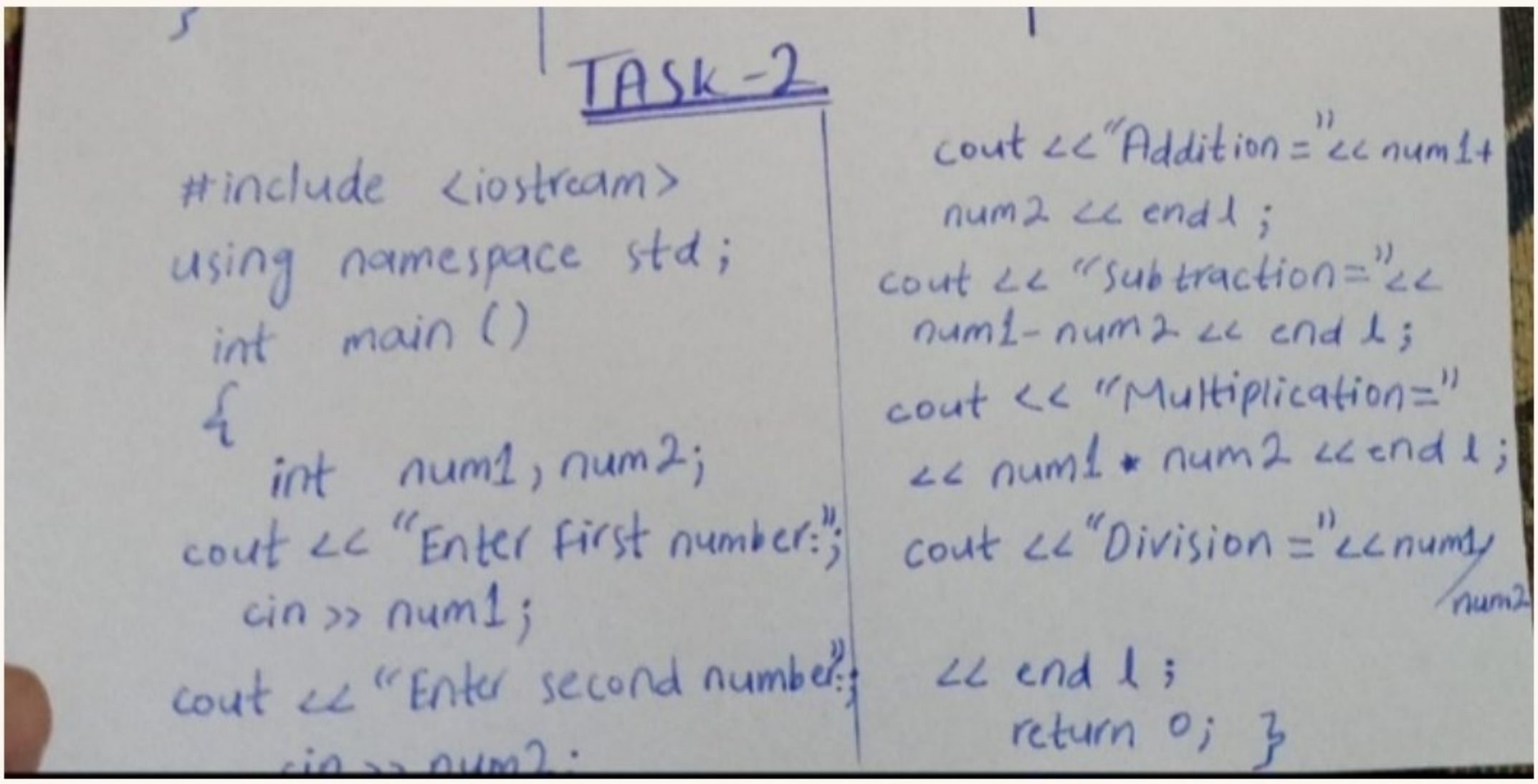
    return 0;
}
```

Output, program 3

```
65
66
```

Lab Task 2

Handwritten code



Typed code

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num1, num2;

    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter second number: ";
    cin >> num2;

    cout << "Addition = " << num1 + num2 << endl;
    cout << "Subtraction = " << num1 - num2 << endl;
    cout << "Multiplication = " << num1 * num2 << endl;
    cout << "Division = " << num1 / num2 << endl;

    return 0;
}
```

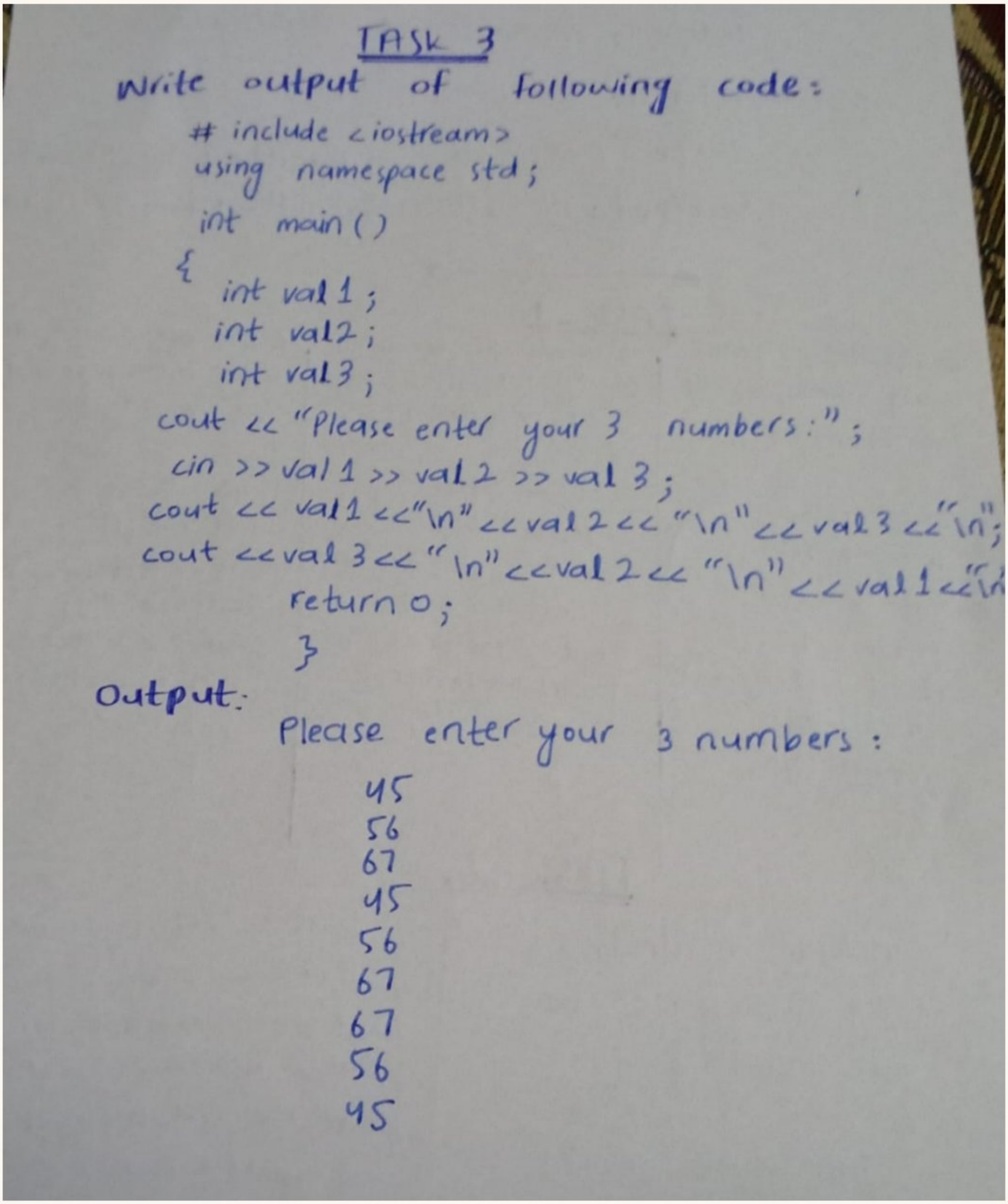
Output

```
Enter first number: 5
Enter second number: 9
Addition = 14
Subtraction = -4
Multiplication = 45
Division = 0

[Program finished]
```

Lab Task 3

Handwritten code



Typed code

```cpp
#include <iostream>
using namespace std;

int main()
{
    int val1;
    int val2;
    int val3;

    cout << "Please enter your 3 numbers:";
    cin >> val1 >> val2 >> val3;

    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";

    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";

    return 0;
}
```

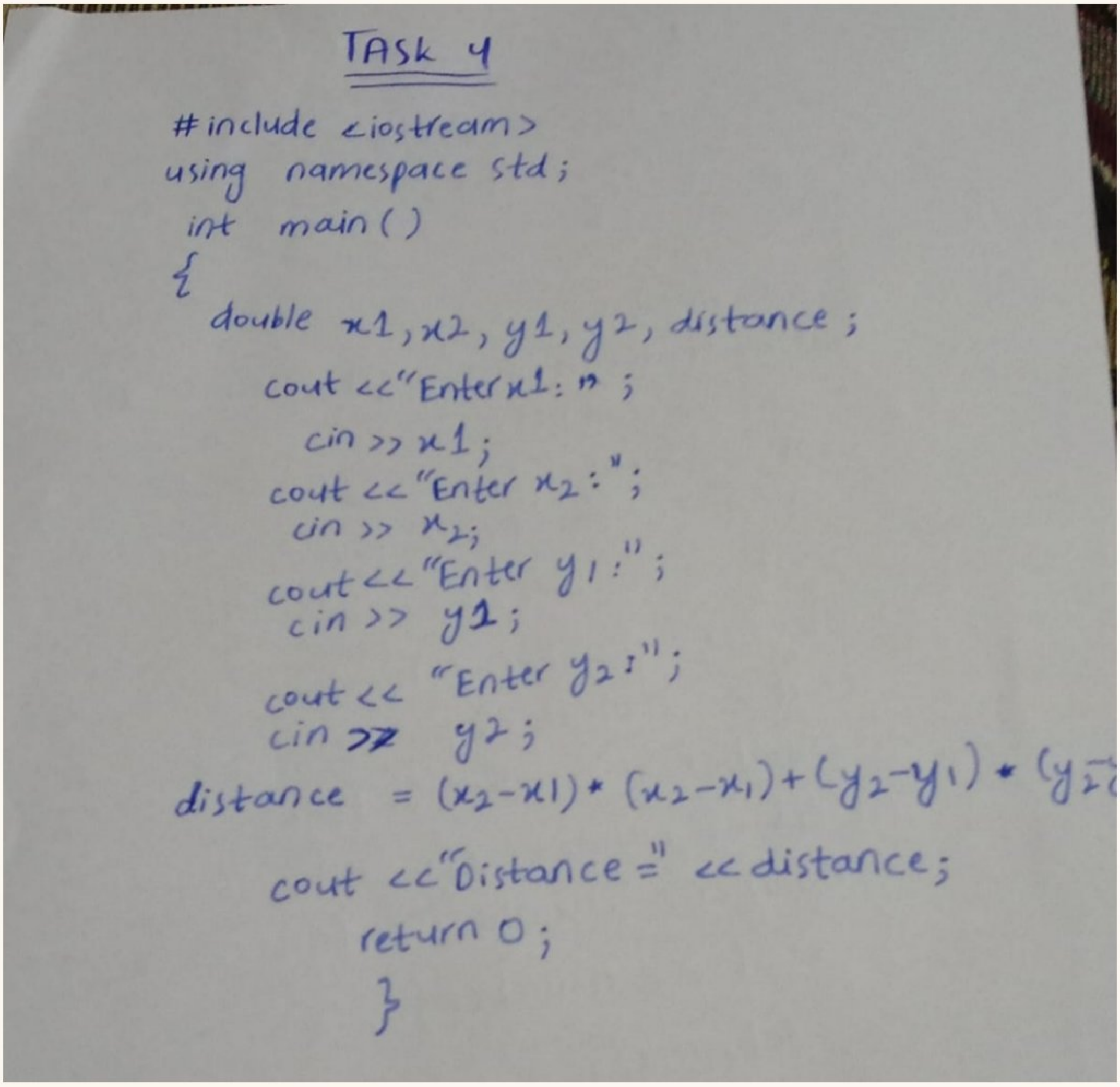
Output

```
Please enter your 3 numbers:45
56
67
45
56
67
67
56
56
45

[Program finished]
```

Lab Task 4

Handwritten code



Typed code

```
#include <iostream>
using namespace std;

int main()
{
    double x1, x2, y1, y2, distance;

    cout << "Enter x1: ";
    cin >> x1;

    cout << "Enter x2: ";
    cin >> x2;

    cout << "Enter y1: ";
    cin >> y1;

    cout << "Enter y2: ";
    cin >> y2;

    distance = (x2 - x1) * (x2 - x1) + (y2 - y1) * (y2 - y1);

    cout << "Distance = " << distance;

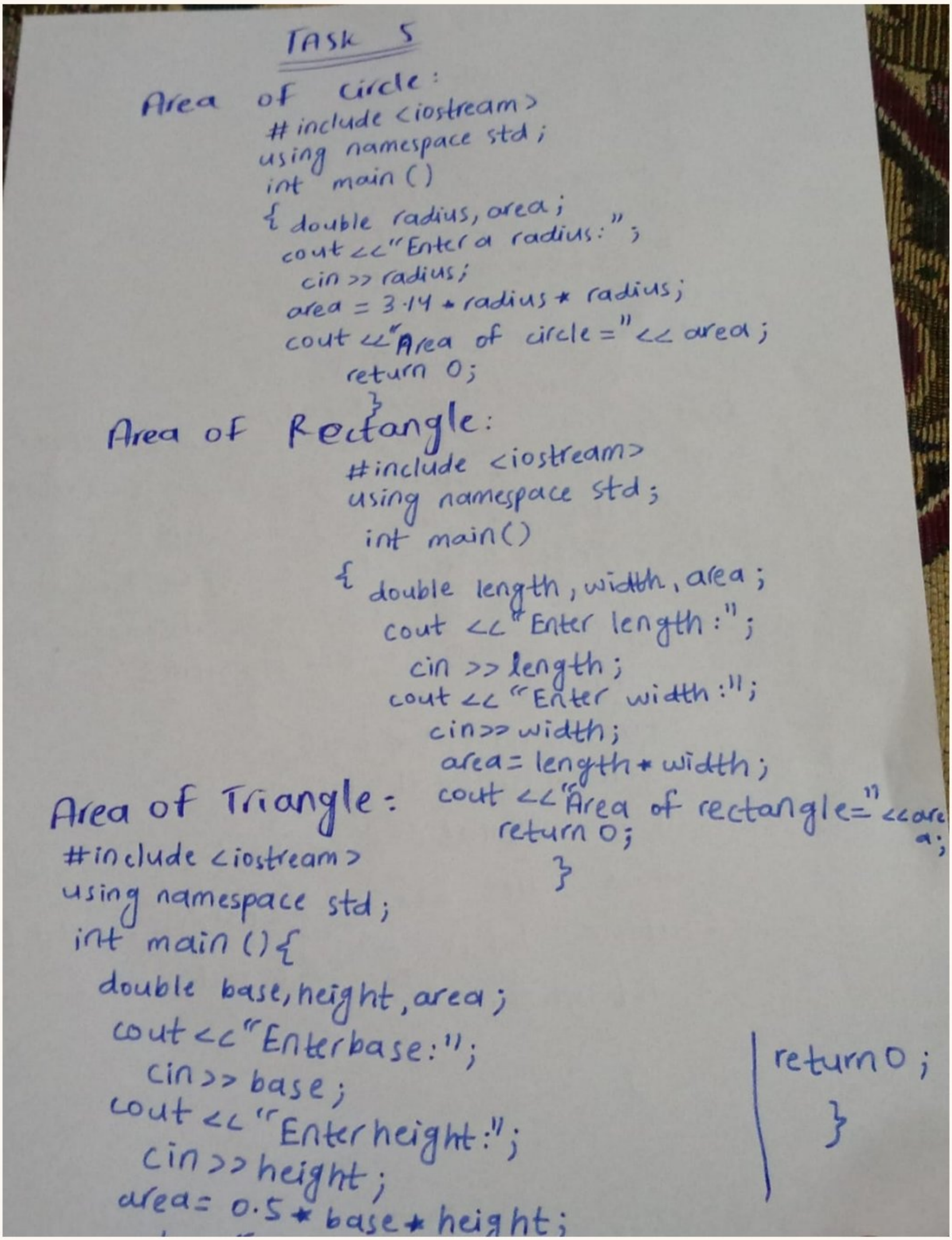
    return 0;
}
```

Output

```
Enter x1: 6
Enter x2: 7
Enter y1: 8
Enter y2: 9
Distance = 2
[Program finished]
```

Lab Task 5

Handwritten code (circle, rectangle and triangle)



Typed code: circle

```cpp
#include <iostream>
using namespace std;

int main()
{
    double radius, area;

    cout << "Enter radius: ";
    cin >> radius;

    area = 3.14 * radius * radius;

    cout << "Area of circle = " << area;

    return 0;
}
```

Output: circle

```
Enter radius: 6
Area of circle = 113.04
[Program finished]
```

Typed code: rectangle

```cpp
#include <iostream>
using namespace std;

int main()
{
    double length, width, area;

    cout << "Enter length: ";
    cin >> length;

    cout << "Enter width: ";
    cin >> width;

    area = length * width;

    cout << "Area of rectangle = " << area;

    return 0;
}
```

Output: rectangle

```
Enter length: 6
Enter width: 7
Area of rectangle = 42
[Program finished]
```

Typed code: triangle

```cpp
#include <iostream>
using namespace std;

int main()
{
    double base, height, area;

    cout << "Enter base: ";
    cin >> base;
    cout << "Enter height: ";
    cin >> height;

    area = 0.5 * base * height;
    cout << "Area of triangle = " << area;

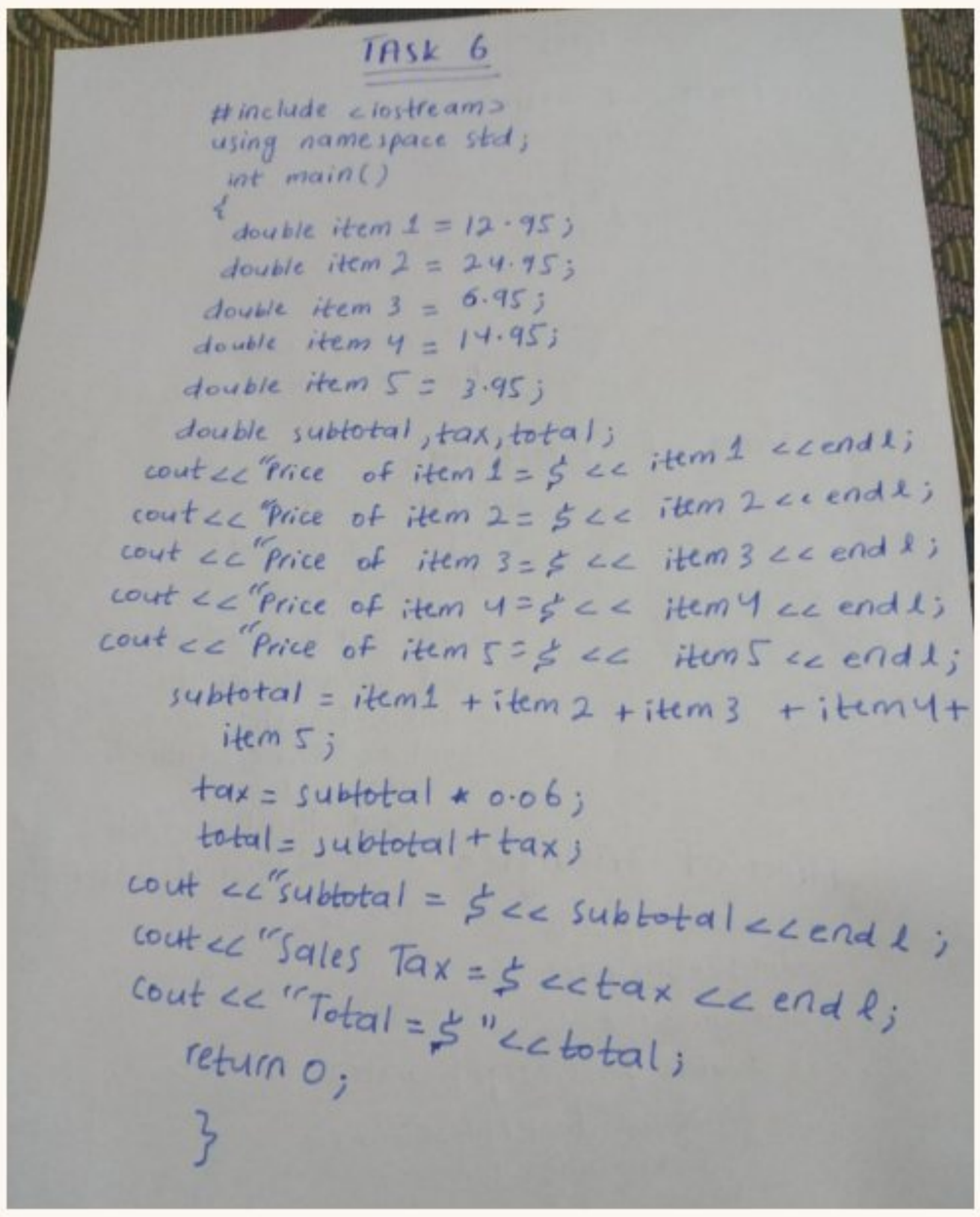
    return 0;
}
```

Output: triangle

```
Enter base: 6
Enter height: 9
Area of triangle = 27
[Program finished]
```

Lab Task 6

Handwritten code



TASK 6

```
#include <iostream>
using namespace std;
 int main()
{
 double item 1 = 12.95;
 double item 2 = 24.95;
double item 3 = 6.95;
double item 4 = 14.95;
double item 5 = 3.95;
double subtotal, tax, total;
cout<<"Price of item 1 = $"<< item 1 <<endl;
cout<<"Price of item 2 = $"<< item 2 <<endl;
cout <<"Price of item 3 = $"<< item 3 <<endl;
cout <<"Price of item 4 = $"<< item 4 <<endl;
cout <<"Price of item 5 = $"<< item 5 <<endl;
   subtotal = item1 + item 2 + item 3 + item 4+
     item 5;
   tax = subtotal * 0.06;
   total = subtotal + tax;
cout <<"subtotal = $"<< subtotal <<endl;
cout<<"Sales Tax = $"<<tax <<endl;
Cout <<"Total = $"<<total;
   return 0;
   }
```

Typed code

```cpp
#include <iostream>
using namespace std;

int main()
{
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;

    double subtotal, tax, total;

    cout << "Price of item 1 = $" << item1 << endl;
    cout << "Price of item 2 = $" << item2 << endl;
    cout << "Price of item 3 = $" << item3 << endl;
    cout << "Price of item 4 = $" << item4 << endl;
    cout << "Price of item 5 = $" << item5 << endl;

    subtotal = item1 + item2 + item3 + item4 + item5;

    tax = subtotal * 0.06;

    total = subtotal + tax;

    cout << "Subtotal = $" << subtotal << endl;
    cout << "Sales Tax = $" << tax << endl;
    cout << "Total = $" << total;

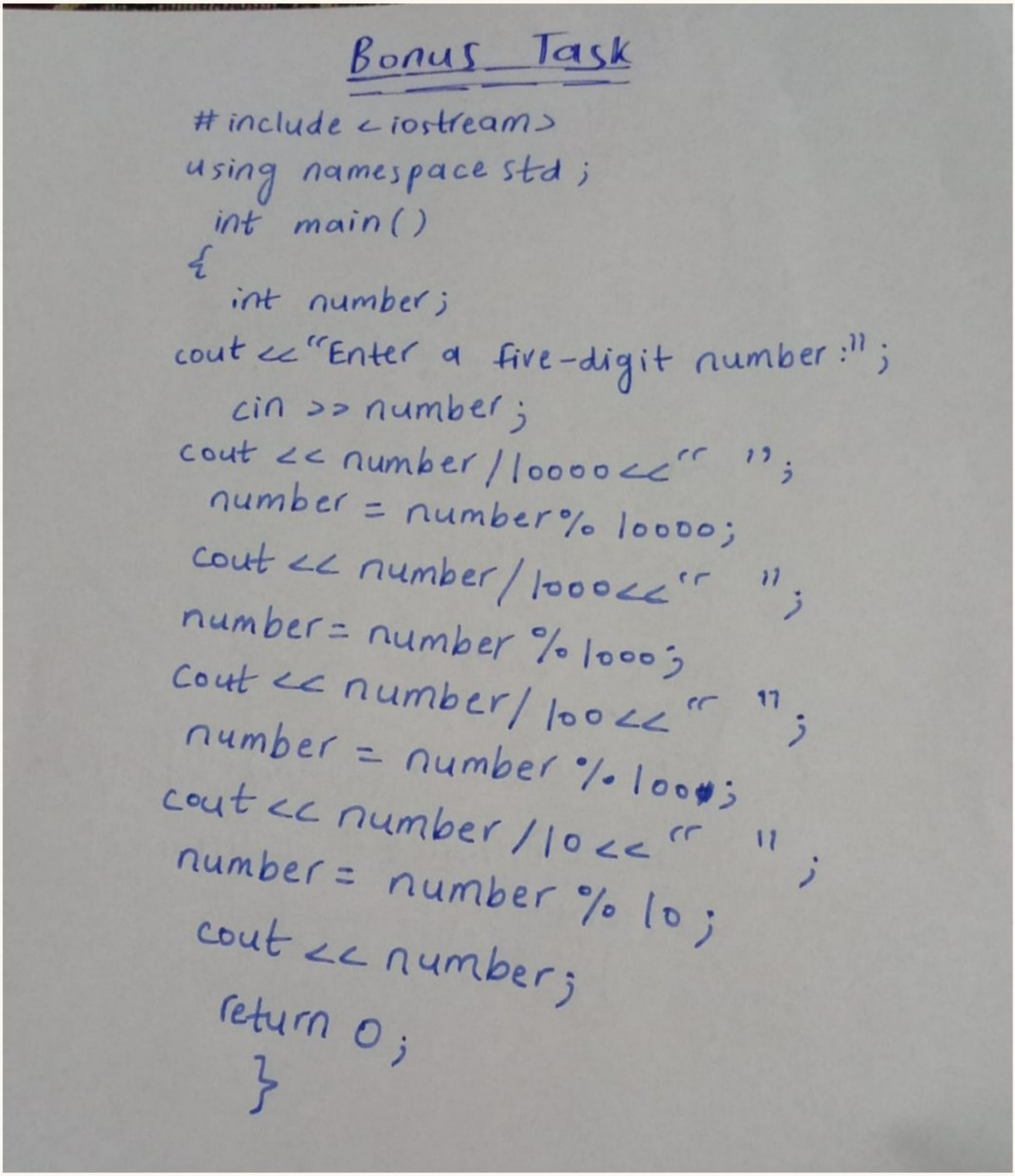
    return 0;
}
```

Output

```
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95
Subtotal = $63.75
Sales Tax = $3.825
Total = $67.575
[Program finished]
```

Bonus Task

Handwritten code



Typed code

```cpp
#include <iostream>
using namespace std;

int main()
{
    int number;

    cout << "Enter a five-digit number: ";
    cin >> number;

    cout << number / 10000 << "    ";
    number = number % 10000;

    cout << number / 1000 << "    ";
    number = number % 1000;

    cout << number / 100 << "    ";
    number = number % 100;

    cout << number / 10 << "    ";
    number = number % 10;

    cout << number;

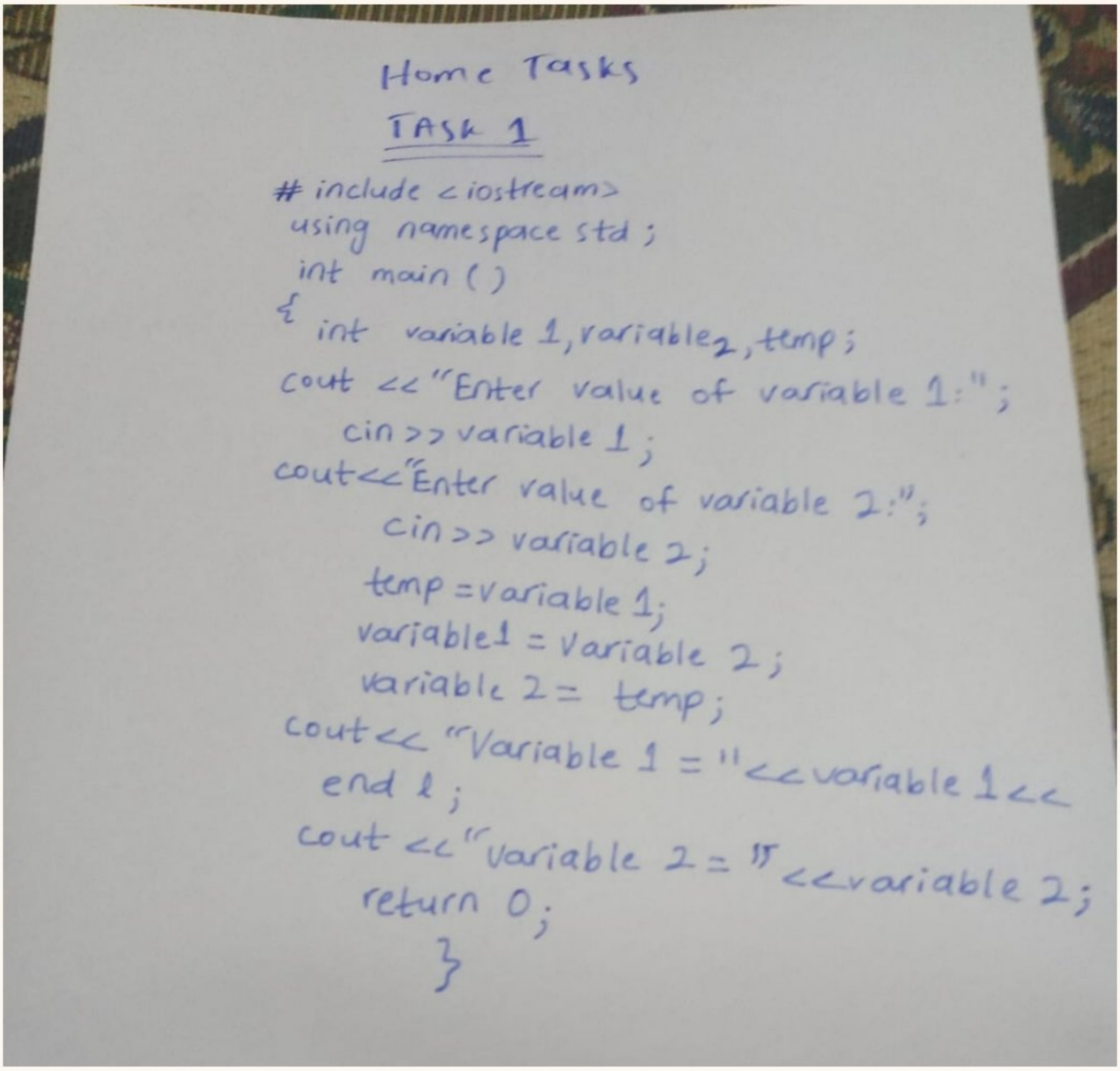
    return 0;
}
```

Output

```
Enter a five-digit number: 4567
0   4   5   6   7
[Program finished]
```

Home Task 1

Handwritten code



Typed code

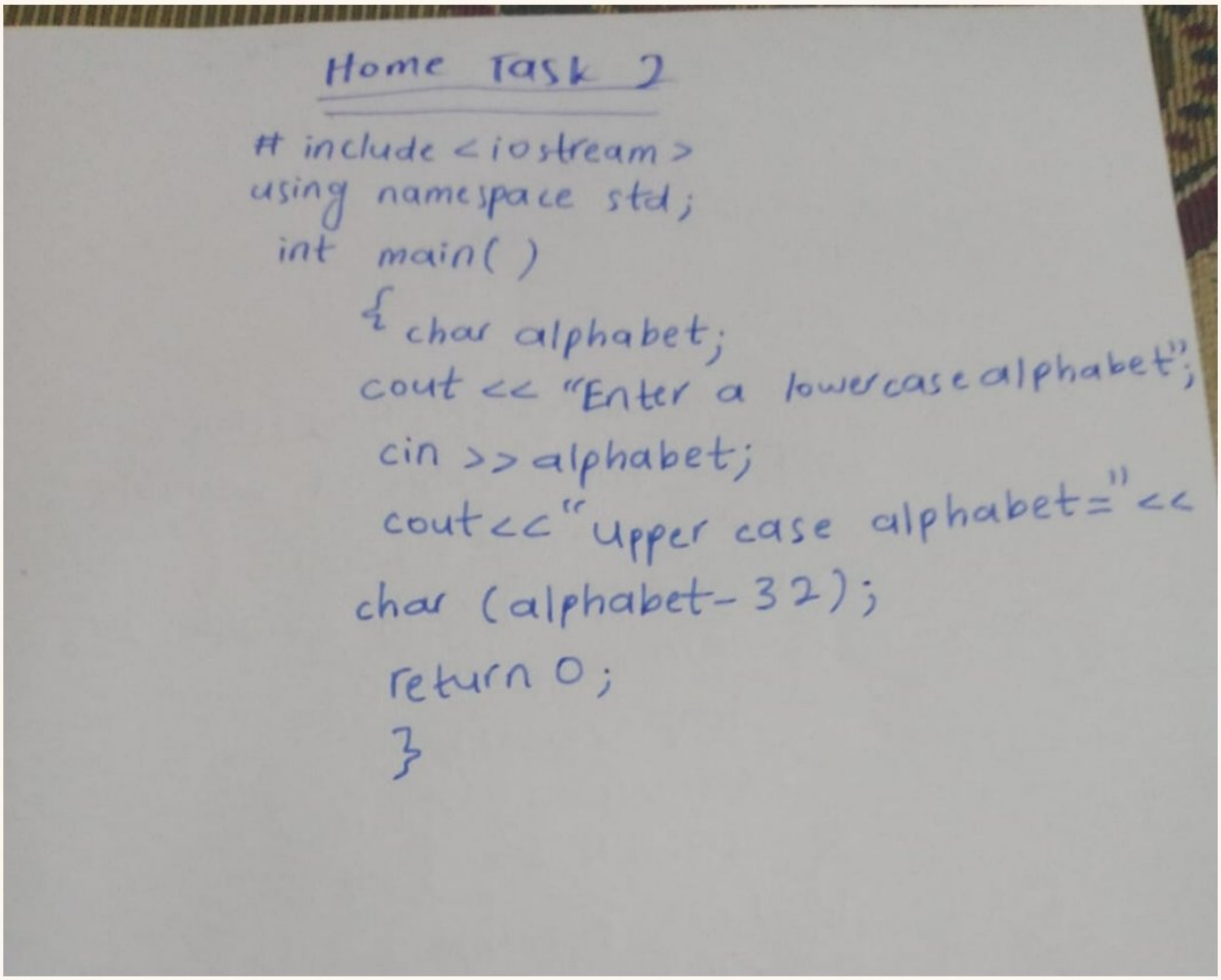
```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int variable1, variable2, temp;
7
8      cout << "Enter value of variable 1: ";
9      cin >> variable1;
10
11     cout << "Enter value of variable 2: ";
12     cin >> variable2;
13
14     temp = variable1;
15     variable1 = variable2;
16     variable2 = temp;
17
18     cout << "Variable 1 = " << variable1 <<
   endl;
19     cout << "Variable 2 = " << variable2;
20
21     return 0;
22 }
```

Output

```
Enter value of variable 1: 6
Enter value of variable 2: 7
Variable 1 = 7
Variable 2 = 6
[Program finished]
```

Home Task 2

Handwritten code



Typed code

```
#include <iostream>
using namespace std;

#include <iostream>
using namespace std;

int main()
{
    char alphabet;

    cout << "Enter a lowercase alphabet: ";
    cin >> alphabet;

    cout << "Uppercase alphabet = " <<
char(alphabet - 32);

    return 0;
}
```

Output

```
Enter a lowercase alphabet=w
upper case alphabet=W
[Program finished]
```

End of Assignment